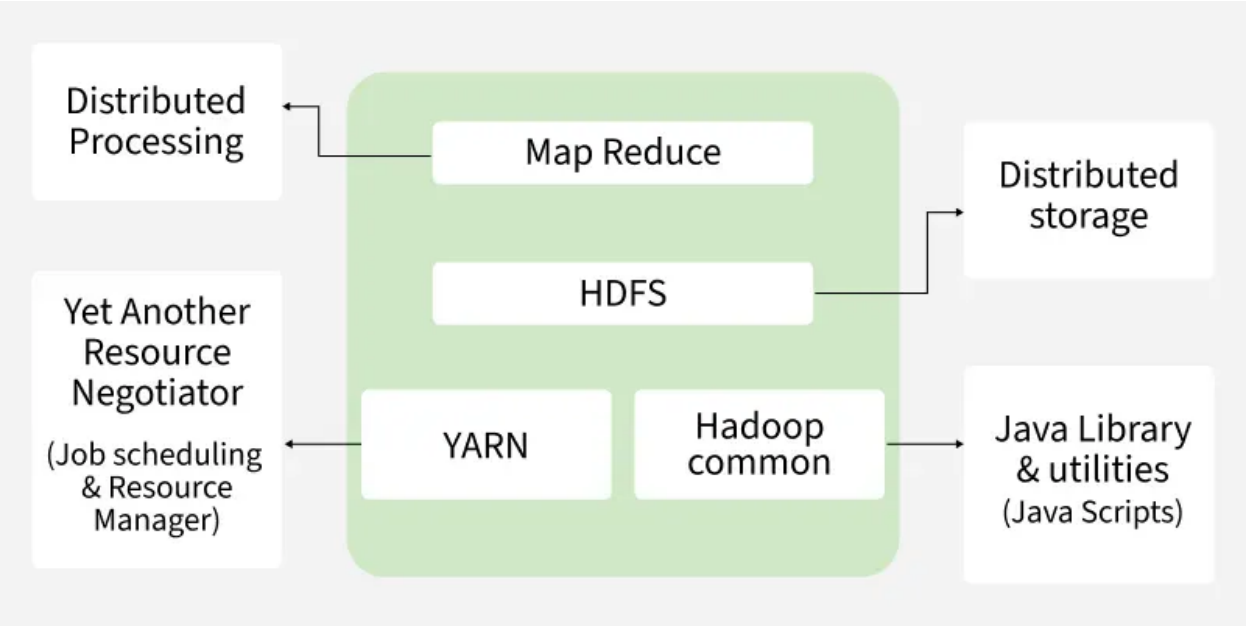
CMP604 – BDA and DV Laboratory

Laboratory 01 - Downloading and installing Hadoop:

Aim: Installation of Hadoop Framework, its components and study the HADOOP ecosystem

Hadoop is a Java-based programming framework that supports the processing and storage of extremely large datasets on a cluster of inexpensive machines. It was the first major open-source project in the big data field and is sponsored by the Apache Software Foundation.

Hadoop is comprised of four main layers:



Hadoop is an open-source framework that allows to store and process big data in a distributed environment across clusters of computers using simple programming models. It is designed to scale up from single servers to thousands of machines, each offering local computation and storage. **Hadoop Architecture:**

The Apache Hadoop framework includes following four modules:

1. **Hadoop Common**: Contains Java libraries and utilities needed by other Hadoop modules. These libraries give file system and OS level abstraction and comprise of the essential Java files and scripts that are required to start Hadoop.
2. **Hadoop Distributed File System (HDFS):** A distributed file-system that provides high-throughput access to application data on the community machines thus providing very high aggregate bandwidth across the cluster.
3. **Hadoop YARN:** A resource-management framework responsible for job scheduling and cluster resource management.
4. **Hadoop MapReduce:** This is a YARN- based programming model for parallel processing of large data sets.

Installation Steps – Linux

Steps to Install Apache Hadoop on Ubuntu 14.04

Step - 1: Install Java (OpenJDK)

Since hadoop is based on java, make sure you have java jdk installed on the system.
Please check the version of java (It should be 1.7 or above)

```
$ java -version
```

If it returns "The program java can be found in the following packages", If Java isn't been installed yet, so execute the following command:

```
$sudo apt-get install default-jdk
```

Step - 2: Configure Apache Hadoop

1. Open bashrc in gedit mode

```
$sudo1. Open gedit bashrc ~/.bashrc file  in geditor
```

2. Set java environment variable

```
export JAVA_HOME=/usr/jdk1.7.0_45/
```

3. Set Hadoop environment variable

```
export HADOOP_HOME=/usr/Hadoop 2.6/
```

4. Apply environment variables

```
$source ~/.bashrc
```

Step - 3: Install eclipse

Step - 4: Copy Hadoop plugins such as

• hadoop-eclipse-kepler-plugin-2.2.0.jar
• hadoop-eclipse-kepler-plugin-2.4.1.jar
• hadoop-eclipse-plugin-2.6.0.jar

Copy these from from **release** folder of **hadoop2x-eclipse-plugin- master to eclipse plugins**

Step - 5: In **eclipse,** start new **MapReduce project**

```
File->new->other->MapReduce project
```

Step - 6: Copy Hadoop packages

Copy packages such as

- commons-io-2.4.jar
- commons-lang3-3.4.jar

Copy these packages in **src** file of MapReduce project

Step - 7: Create Mapper, Reducer, and driver

```
Inside a project->src->File->new->other->Mapper/Reducer/Driver
```

Step - 8: Copy Log file

Copy **log4j.properties** from **src** file of hadoop in src file of MapReduce project

Hadoop is powerful because it is extensible and it is easy to integrate with any component. Its popularity is due in part to its ability to store, analyze and access large amounts of data, quickly and cost effectively across clusters of commodity hardware. Apache Hadoop is not actually a single product but instead a collection of several components. When all these components are merged, it makes Hadoop very user friendly.

Installation Steps – Windows

We will install Hadoop in stand-alone mode and run one of the example MapReduce programs it includes to verify the installation.

Step – 1: Prerequisites

- **Java JDK:** Download and install Java Development Kit (JDK 8 or newer).

  Download and install Java JDK 8 or 11. Avoid installing in a path with spaces (e.g., use C:\Java\jdk1.8.0 instead of C:\Program Files\Java) to prevent configuration errors.
- **7-Zip:** For unzipping .tar.gz files.
- **Winutils.exe:** Download the Hadoop native binaries for Windows (search "hadoop winutils github"). Download version-specific winutils.exe and hadoop.dll from a trusted repository like cdarlint/winutils.

Step – 2: Download & Extract Hadoop

- Download a stable Hadoop version (e.g., 3.x) from the Apache website.
- Download a stable release (e.g., 3.3.x or 3.4.x) from the Official Apache Hadoop Releases
- Use 7-Zip (run as admin) to extract the Hadoop .tar.gz file to C:\hadoop (rename the extracted folder).

1. **Extract Hadoop:** Unzip the downloaded tar file to a simple directory like C:\Hadoop.
2. **Apply Windows Fix:** Copy the downloaded winutils.exe and hadoop.dll into your C:\Hadoop\bin folder.
3. **Configure Variables:**
   - Go to **System Properties > Environment Variables**.
   - Create JAVA_HOME pointing to your JDK folder (e.g., C:\Java\jdk1.8.0).
   - Create HADOOP_HOME pointing to your Hadoop folder (e.g., C:\Hadoop).
   - Edit the **Path** variable and add:
     - %JAVA_HOME%\bin
     - %HADOOP_HOME%\bin
     - %HADOOP_HOME%\sbin.

Step – 3: Set Environment Variables

- **JAVA_HOME**: C:\Program Files\Java\jdk-XX (your JDK path).
- **HADOOP_HOME**: C:\hadoop (your Hadoop path).
- Add %HADOOP_HOME%\bin and %HADOOP_HOME%\sbin to your system's Path variable.

Step – 4: Configure Hadoop Files (in %HADOOP_HOME%\etc\hadoop)

- **core-site.xml**: Set fs.defaultFS to `hdfs://localhost:9000`.
- **hdfs-site.xml**: Define dfs.replication (e.g., 1) and create data directories (e.g., C:\hadoop\data\namenode, C:\hadoop\data\datanode).
- **mapred-site.xml**: Set mapreduce.framework.name to yarn.
- **yarn-site.xml**: Add <property><name>yarn.nodemanager.aux services</name><value>mapreduce_shuffle</value></property>

Edit the configuration files located in %HADOOP_HOME%\etc\hadoop.

- Modify core-site.xml to set the default filesystem, for example, to `hdfs://localhost:9000`.
- Edit hdfs-site.xml to define data replication and the paths for namenode and datanode data directories. Ensure these directories are created manually first. For instance, set replication to 1 and define directories `like file:///C:/Hadoop/data/namenode` and `file:///C:/Hadoop/data/datanode`.
- In `mapred-site.xml`, set the MapReduce framework name to yarn.
- In `hadoop-env.cmd`, explicitly set the `JAVA_HOME` variable to your JDK path. You can find examples of the XML configurations in the referenced sources.

Step - 5: Add winutils.exe

- Copy the winutils.exe (and Hadoop native binaries) into %HADOOP_HOME%\bin.

Step – 6: Start Hadoop

- Open Command Prompt (as Admin).
- Format the NameNode (only once): hadoop namenode -format.
- Start services:
  - start-dfs.cmd
  - start-yarn.cmd
- Check the Resource Manager at http://localhost:8088.

1. Open Command Prompt as Administrator and then run `hdfs namenode -format`.
2. Execute `start-all.cmd` (or `start-dfs.cmd` and `start-yarn.cmd` separately) from the `sbin` folder.
3. Check the Web UIs at `http://localhost:9870` for the NameNode and `http://localhost:8088` for the ResourceManager.